Pretrained 3D Foundation Models for Airfoil Surrogate Modeling

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***Abstract*— Surrogate models are essential for accelerating aerodynamic design, yet they remain limited by the high cost of generating CFD training data. This research proposes a decoupled learning strategy that utilizes pretrained 3D foundation models, PointNet and PointMAE, to provide general-purpose geometric features, reducing the reliance on expensive physical labels. By transferring knowledge from the large-scale ShapeNet dataset to the 2D airfoil domain, we investigate the impact of pretraining objectives and geometric alignment on model performance. Our results indicate that reconstruction-based pretraining, combined with 3D extrusion, a process of adding volumetric depth to 2D airfoil contours to align them with the 3D pretraining distribution, yields performance gains of up to 35%. This demonstrates that 3D foundation models can effectively bridge domain gaps and enhance surrogate accuracy in data-scarce regimes.**

I. Introduction

The high computational cost of Computational Fluid Dynamics (CFD) remains a primary bottleneck in aerodynamic design, where iterative optimization requires evaluating thousands of geometric configurations. While machine learning surrogate models offer near-instantaneous inference, their training remains dependent on large-scale CFD datasets to map geometries to physical fields.

This study addresses this data dependency by decoupling geometric representation from physical mapping. We hypothesize that because geometric understanding does not require physical labels, it can be offloaded to 3D foundation models pretrained on massive, non-aerodynamic datasets. Specifically, we utilize PointNet and PointMAE as frozen encoders to extract global features from airfoil point clouds. A central challenge in this transfer learning approach is the domain gap between the isotropic, volumetric objects of the pretraining set (ShapeNet) and the anisotropic, 2D nature of airfoils.

To bridge this gap, we evaluate various geometric alignment strategies to project airfoil data into a distribution more familiar to the foundation models. Furthermore, we analyze how the pretraining objective influences feature utility, comparing PointNet's high-level semantic features against PointMAE's low-level structural reconstructions. Our findings provide a framework for integrating general-purpose 3D foundation models into specialized physical surrogate pipelines, significantly improving predictive performance without increasing the CFD data requirement.

II. Methodology and System Architecture

To evaluate the utility of 3D foundation models in aerodynamic surrogate modeling, we implement an extended pipeline designed to decouple geometric feature extraction from physical field prediction. The architecture comprises three components represented in figure 1: a pretrained 3D foundation model, a fusion mechanism, and a downstream surrogate model:

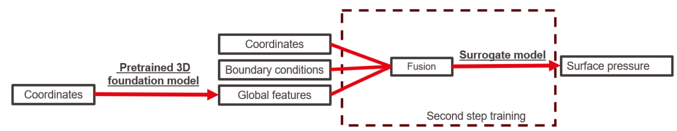

Fig. 1. Extended pipeline representation

A. Pretrained 3D Foundation Models

We investigate two distinct 3D foundation models as feature extractors. Both are pretrained on the ShapeNet dataset but differ fundamentally in their architecture and learning objectives:

- **PointNet:** A supervised model pretrained for shape classification. It employs shared Multi-Layer Perceptrons (MLPs) to extract point-wise features, which are aggregated into a global feature vector via a max-pooling operation. This objective encourages the model to learn high-level semantic representations by identifying "critical points" that define an object's category. [1]
- **PointMAE:** A self-supervised model pretrained for geometric reconstruction. The architecture utilizes a Transformer-based autoencoder to recover masked point patches from the input. This task necessitates a low-level understanding of surface continuity and spatial relationships. [2]

In this framework, the foundation models are kept frozen. By serving as static feature extractors, we ensure that performance gains are attributable to the transferability of pretrained geometric knowledge rather than new information learned from the CFD-labeled training set.

B. Surrogate Model Architectures

The extracted features are benchmarked across four baseline architectures adapted from the AirfRANS dataset [3]:

- **MLP:** A standard multi-layer perceptron that processes points independently.

- **GraphSAGE:** A graph neural network utilizing inductive message passing to capture local neighborhood information [4].
- **Graph U-Net:** A hierarchical graph architecture using pooling and unpooling layers to capture multi-scale geometric structures [5].
- **PointNet:** A specific implementation where the predictor follows the PointNet segmentation head, allowing for a study of architectural redundancy [1].

While the original AirfRANS framework predicts a comprehensive set of flow variables, including velocity, pressure, and turbulent viscosity across the entire fluid domain, this study narrows the scope to surface pressure prediction. This reduction significantly decreases computational overhead and training time, while maintaining a robust proxy for evaluating the surrogate models' ability to capture critical aerodynamic characteristics.

C. Adaptive Fusion and Gating Mechanism

A critical component of the architecture is the integration of the global geometric features (F_{global}) with local point features (F_{local}), which include point coordinates and boundary conditions. We employ a learned gating mechanism to adaptively modulate the influence of the global context based on the local geometric requirements of each point.

The fusion process is formulated as follows:

- **Gate Generation:** Local features are passed through a shallow MLP (8,8) with a sigmoid activation function to produce a point-wise gate:
$$G = \sigma(\text{MLP}(F_{local}))$$
- **Global Feature Integration:** The global feature vector is broadcast to all points and modulated by the gate. The final fused representation (F_{fused}) is the sum of the local features and the gated global features:
$$F_{fused} = F_{local} + (G \odot F_{global})$$

This gated summation allows the surrogate model to prioritize global information in regions where local descriptors are insufficient for accurate pressure prediction. The resulting F_{fused} is subsequently processed by the surrogate model to predict the surface pressure.

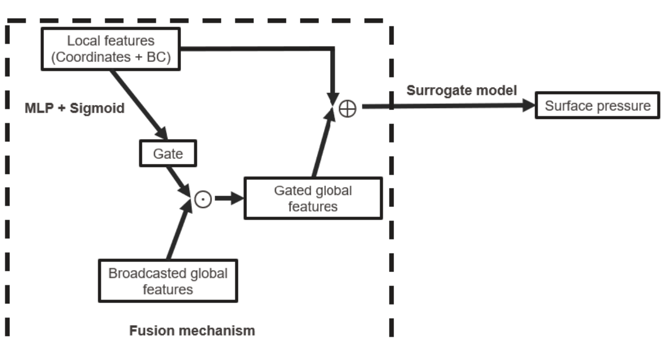

Fig. 2. Fusion mechanism representation

III. Latent Space Representation Analysis

To assess whether pretrained 3D foundation models capture meaningful geometric information for aerodynamic shapes, we perform a qualitative analysis of the extracted global feature vectors. Given the high dimensionality of these vectors, we employ t-Distributed Stochastic Neighbor Embedding (t-SNE) [6] to project the latent space into a two-dimensional plane while preserving local similarity relationships.

A. Dimensionality Reduction and Visualization

We extract the global feature vectors for the entire airfoil dataset using both frozen PointNet and PointMAE encoders. The t-SNE algorithm [6] is then applied to these vectors, converting Euclidean distances in the high-dimensional space into conditional probabilities that represent similarities. By minimizing the Kullback-Leibler (KL) divergence between the high-dimensional and low-dimensional distributions, t-SNE produces a map where airfoils with similar geometric properties are clustered together.

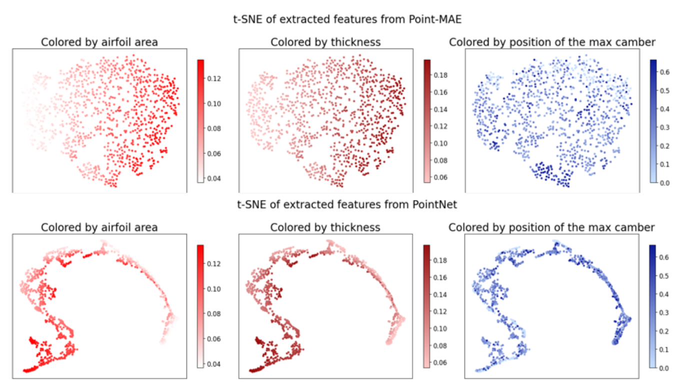

Fig. 3. t-SNE visualization of the latent spaces. Each point represents one airfoil, and its position corresponds to the global feature vector extracted by the pretrained model.The colors indicate known geometric properties that are meaningful for airfoil geometries: airfoil area, thickness, and position of the maximum camber.

To validate the physical relevance of the latent space, we color the t-SNE embeddings based on known geometric descriptors: airfoil area, maximum thickness, and the longitudinal position of the maximum camber.

1) Geometric Property Characterization

Figure 3 results reveal clear, smooth color gradients. Specifically, PointMAE exhibits a gradient from left to right, while PointNet shows a similar, though more constrained, right-to-left gradient. The presence of these gradients confirms that despite being trained on unrelated 3D objects like furniture and vehicles, the models have learned spatial priors that inherently encode the fundamental geometric variations of airfoils.

2) Comparison of Pretraining Objectives

A significant structural difference is observed between the two latent spaces. The PointMAE embedding is more uniformly distributed and covers a broader area, whereas the PointNet embedding appears more irregular and clustered.

We attribute this to the models' respective pretraining objectives:

- **PointNet classification model:** As a classification-oriented model, PointNet is trained to discard intra-class variations in favor of discriminative features. It maps shapes to a high-level semantic space, which may compress fine geometric nuances that do not contribute to category identification.
- **PointMAE reconstruction model:** The reconstruction task forces PointMAE to preserve a detailed geometric description of the input to recover masked patches. This results in a latent space that is more sensitive to low-level structural variations, such as subtle changes in surface curvature, which are critical for characterizing aerodynamic performance.

This analysis suggests that while both models capture relevant features, the reconstruction-based approach of PointMAE provides a more descriptive and continuous representation of the airfoil manifold, potentially offering a superior foundation for downstream physical field prediction.

IV. Domain Adaptation through Geometric Alignment

A significant challenge in utilizing 3D foundation models for airfoil analysis is the geometric domain gap. ShapeNet consists of isotropic, volumetric 3D objects, while the AirfRANS dataset comprises anisotropic, 2D contours. To maximize the use of the 3D foundation models, we evaluate several alignment strategies designed to project airfoil data into a spatial distribution more familiar to the foundation models pretraining dataset.

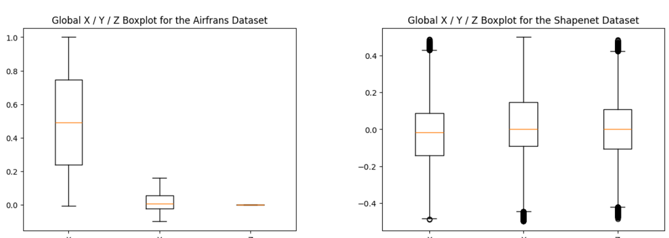

Fig. 4. Statistic comparison between Shapenet and Airfrans

A. 2D transformations

We begin with simple 2D transformations, where the airfoils are centered and standardized.

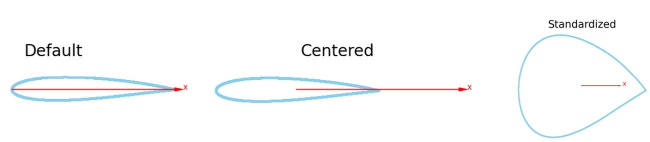

Fig. 5. 2D transformations

B. Manual 3D Geometric Transformations

To address the topological mismatch between 2D contours and 3D volumes, we also explore methods to introduce volumetric structure.

1) 3D Extrusion

To create the upper and lower surfaces, we compute the convex hull of the airfoil and uniformly sample points within this polygon at fixed depths of $-h/2$ and $h/2$. The lateral surfaces are generated by sampling points directly from the airfoil contour and assigning them random depths within the thickness range $[-h/2, h/2]$. This ensures a continuous volumetric manifold rather than disjoint planes.

2) Torus Embedding

We also evaluate a revolution-based embedding where the 2D airfoil is revolved around the vertical axis to create a torus-like surface. The airfoil is shifted by a fixed major radius along the x-axis, and each point is rotated by a randomly sampled angle. This random sampling introduces noise that prevents structured artifacts, such as perfectly aligned rings, resulting in a more uniformly distributed point cloud on the manifold.

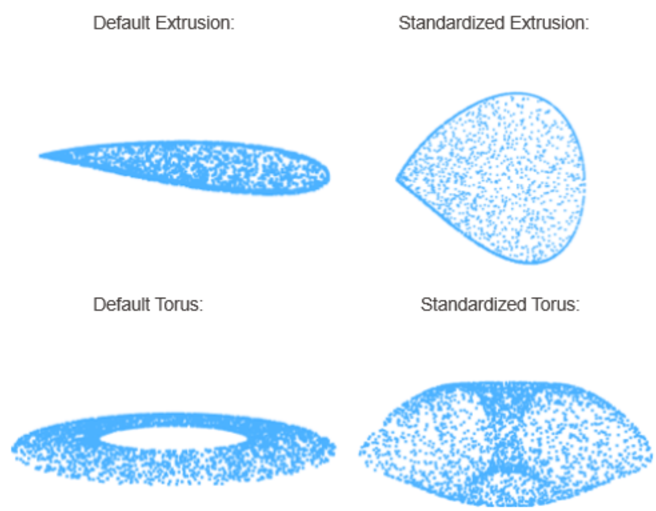

Fig. 6. 3D transformations

C. Automated Alignment via Optimal Transport

Beyond manual transformations, we investigated an automated approach to map airfoil distributions onto the ShapeNet distribution using Optimal Transport (OT) [7], [8].

- **Distribution Approximation:** We fit a Gaussian Mixture Model (GMM) to the ShapeNet point distribution to capture its multi-modal density.
- **Reference Sampling:** A fixed reference point cloud is sampled from the GMM to serve as the target distribution.
- **Optimal Transport Mapping:** Each airfoil point cloud is interpreted as a discrete probability distribution. We solve the entropic-regularized OT problem using the Sinkhorn algorithm to find a transport plan that minimizes the total

cost (squared Euclidean distance) of moving mass from the airfoil points to the reference cloud.
- **Barycentric Projection:** As mass from a single source point can be split across multiple target points, we apply a barycentric projection to map each original airfoil point onto the reference support.

Theses four steps are presented in Figure 7:

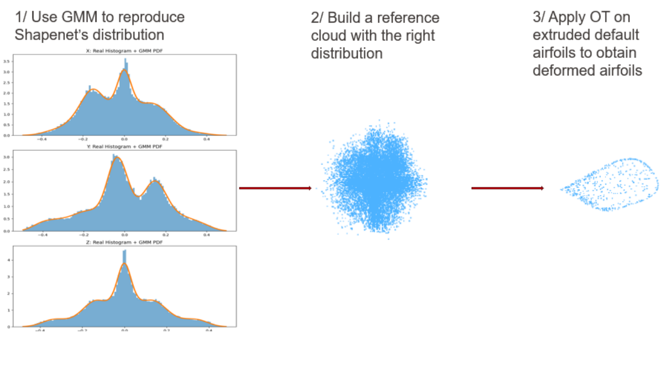

Fig. 7. OT transformation steps

In practice, the learned transformation did not yield significant additional value. The resulting geometries were similar to simple extruded airfoils, often collapsing the fine aerodynamic features when mass conservation forced transport to distant points in the subsampled target. Consequently, we focus on the more robust manual 3D extrusion for the remainder of the analysis.

V. Experimental Results

The effectiveness of our decoupled framework is evaluated by measuring the reduction in Mean Squared Error (MSE) for surface pressure prediction across several surrogate backbones:

$$\text{Relative MSE reduction (\%)} = \frac{\text{MSE}_{\text{baseline}} - \text{MSE}_{\text{global}}}{\text{MSE}_{\text{baseline}}} \times 100$$

A. Architectural Redundancy in PointNet

We first examine the impact of adding foundation features to the surrogate models using the default airfoil representation.

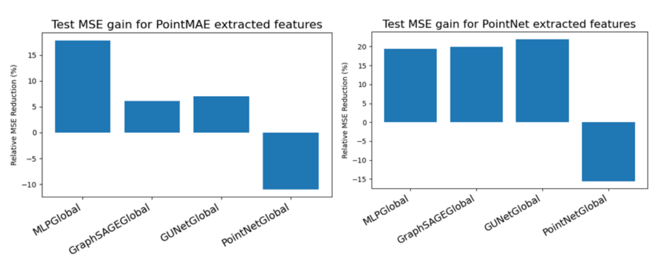

Fig. 8. Impact of the extracted features on the surrogate models performances

Figure 8 shows that, unlike other surrogate models, the PointNet surrogate does not exhibit a significant improvement in test loss when augmented with external global features.

This result is expected, as the PointNet architecture inherently performs global feature extraction. The surrogate model follows a segmentation-style branch: after extracting a global feature through shared MLPs and a max-pooling operation, this feature is broadcast to all points and concatenated with local descriptors. The only difference between PointNet used as a foundation model and PointNet used as a surrogate lies in the task-specific layers following feature extraction. Consequently, adding external global features is largely redundant. For analyzing the impact of domain reduction on the performances, we therefore focus on the MLP, GraphSAGE, and Graph U-Net architectures.

B. Comparison of Geometric Alignment Strategies

We evaluate a wide range of feature extraction strategies across both 2D and 3D domains. Again, performance is reported as relative gain compared to the baseline, where higher values indicate superior pressure field prediction.

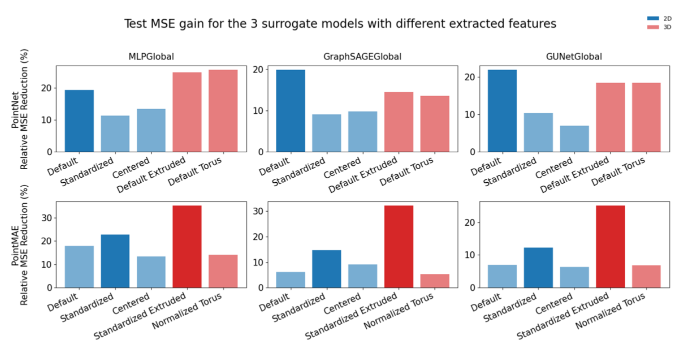

Fig. 9. Test MSE improvements for MLP, GraphSAGE, and Graph U-Net using different extracted features.

1) 2D Transformations

First we focus on the 2D transformations (blue bars in Figure 9). The two foundation models exhibit divergent behavior. For PointNet, the default airfoil representation consistently yields the best results; while 2D transformations degrade the quality of the extracted features. Conversely, for PointMAE, standardizing the airfoil geometry leads to improved performance.

Based on these 2D findings, we apply 3D transformations only to the optimal 2D cases: the default representation for PointNet and the standardized representation for PointMAE.

2) 3D Transformations

Analyzing the red bars of Figure 9, we observe that the impact of 3D alignment further highlights the difference between the encoders:

- PointNet: Still shows no consistent improvement with 3D transformations.
- PointMAE: Benefits significantly from 3D extrusion, with performance gains reaching up to 35% for the MLP.
- Torus Embeddings consistently yield weaker results compared to simple extrusion, suggesting that the revolution-based transformation may introduce geometric distortions that blur important aerodynamic information.

VI. Discussions

A. Why PointNet suffers from Alignment

The reduced performance of PointNet under geometric transformations can be attributed to its classification-based pretraining and the reliance on the max-pooling operation.

- First, when a geometric transformation only slightly modifies the airfoil, at a high semantic level, the shape is still recognized as an airfoil, which is sufficient for a classification-trained model.
- In addition, we know that PointNet global features are determined by a sparse subset of critical points. Therefore after small transformations, the point cloud can remain within the same learned upper-bound shape, so the same critical points dominate the max-pooling operation and the global feature vector stays essentially unchanged.

Finally, more aggressive transformations like Torus embedding may even hurt performance by artificially changing which points become ”critical,” thus diluting the geometric information.

B. Why PointMAE Benefits from Alignment

In contrast, PointMAE is trained for reconstruction using a patch-based masked autoencoder with self-attention. It explicitly learns spatial relationships and global geometric structure at a low level. In this context, geometric transformations serve three vital roles:

- Amplification: They amplify meaningful geometric differences between distinct airfoil profiles.
- Domain Bridging: They reduce the domain gap between the target airfoils and the ShapeNet objects seen during pretraining.
- Volumetric Structure: They introduce the 3D manifold structure (depth and surface continuity) that allows the Transformer-based patches to resolve the geometry more effectively.

VII. Conclusion

In this work, we investigated whether pretrained 3D foundation models can improve aerodynamic surrogate modeling in regimes where CFD data is limited. We demonstrated that global geometric features extracted from these models can indeed enhance surrogate performance, despite the significant discrepancy between the pretraining domain (ShapeNet) and the target airfoil geometries. This finding underscores that pretrained 3D models learn transferable geometric representations that are applicable to specialized engineering domains.

Our results also highlight that the utility of global features is highly dependent on the specific 3D foundation model utilized. While PointNet shows little to no improvement with geometric transformations, PointMAE benefits significantly from geometric alignment and 3D extrusion. We attribute this divergence both to architectural choices, specifically PointNet's max-pooling invariance versus PointMAE's patch-based attention, and to the models' respective pretraining objectives of classification versus reconstruction.

Finally, we acknowledge several limitations of this study. Feature quality was evaluated indirectly through downstream surrogate performance, and the global features were integrated using relatively simple fusion mechanisms. This suggests that the performance gains observed in this work likely represent a lower bound. Future research exploring more advanced fusion strategies could further enhance the ability of surrogate models to leverage these rich geometric priors.

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